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Research Article

BrainNet: Precision Brain Tumor Classification with Optimized EfficientNet Architecture

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Brain tumors significantly impact human health due to their complexity and the challenges in early detection and treatment. Accurate diagnosis is crucial for effective intervention, but existing methods often suffer from limitations in accuracy and efficiency. To address these challenges, this study presents a novel deep learning (DL) approach utilizing the EfficientNet family for enhanced brain tumor classification and detection. Leveraging a comprehensive dataset of 3064 T1-weighted CE MRI images, our methodology incorporates advanced preprocessing and augmentation techniques to optimize model performance. The experiments demonstrate that EfficientNetB(07) achieved 99.14%, 98.76%, 99.07%, 99.69%, 99.07%, 98.76%, 98.76%, and 99.07% accuracy, respectively. The pinnacle of our research is the EfficientNetB3 model, which demonstrated exceptional performance with an accuracy rate of 99.69%. This performance surpasses many existing state-of-the-art (SOTA) techniques, underscoring the efficacy of our approach. The precision of our high-accuracy DL model promises to improve diagnostic reliability and speed in clinical settings, facilitating earlier and more effective treatment strategies. Our findings suggest significant potential for improving patient outcomes in brain tumor diagnosis.

1. Introduction

Brain tumors are lumps or abnormal growths of cells seen in the brain or central nervous system [1]. These tumors can be benign (noncancerous), malignant (cancerous), or metastatic (spreading from another region of the body to the brain) [2]. They can also be primary brain tumors or secondary brain tumors. Primary brain tumors develop in the brain and can be further divided into a variety of forms, including meningiomas, pituitary adenomas, gliomas (including glioblastoma and astrocytoma), and others [3, 4]. Metastatic brain tumors are tumors that have moved from other regions of the body to the brain, frequently from malignancies like melanoma, breast, or the lungs [5, 6]. Most brain tumors have an unknown specific source. However, some risk factors, such as ionizing radiation exposure, a family history of brain tumors, some genetic disorders, and sporadically some environmental exposures, might raise the risk of having a brain tumor [7]. The choice of treatment depends on the type of brain tumor, its location, and the overall health of the patient [8]. Treatment plans are often developed in consultation with a team of medical specialists, including neurosurgeons, oncologists, and radiation therapists [9–11]. The prognosis for brain tumors can vary significantly depending on the type of tumor, its stage, the patient's age, and general health [12]. While some brain tumors can be successfully managed or treated, others might be more difficult [13]. The prognosis for those with brain tumors can be improved with early detection and prompt treatment [14]. About 771,110 cases of central nervous system (CNS) neoplasms will be diagnosed worldwide in the next five years. All central nervous system tumors have a global incidence of 3.9 per 100,000 person-years for primary brain tumors, which make up 1.7% of all cancers [15]. This incidence varies by age, gender, race, and region, with Northern Europe having the highest frequency, followed by Australia, the United States, and Canada [16].

Deep learning (DL) is primarily concerned with neural networks that mimic how the human brain functions by processing and learning from data using multiple layers of interconnected nodes or synthetic neurons [17–19]. The reason it has many layers (deep neural networks) and can automatically learn and extract features from raw data is why it is called "deep" learning. Here is why, DL is useful for brain tumor classification [20]. In feature extraction, DL models can automatically learn and extract relevant features from raw image data. These characteristics may include the tumor's shape, texture, and other traits in the classification of brain tumors [21]. Traditional machine learning (ML) techniques might call for labor-intensive, inefficient feature extraction that is done by hand [22]. DL models are able to spot subtle, complex patterns in the data in complex patterns, which can be essential for classifying various types of brain tumors or determining their malignancy [23].

Fine-tuning a DL model that has already been trained for a particular task, in this case, classifying brain tumors, is known as transfer learning [24]. Because the model has already learned general features and can be tailored to your particular problem, this can save a significant amount of time and computational resources [25]. In scalability, DL models can be scaled to handle large datasets, which makes them suitable for medical applications where you might need to analyze thousands of medical images [26]. In classifying brain tumors, DL can assist in automating the process of identifying and categorizing tumors from medical images, making it quicker and possibly more accurate than manual assessment. DL models are an effective diagnostic tool that can help radiologists and other medical professionals [27].

In neuroscientific research, early-stage detection and categorization of brain tumors cannot be overstated due to their direct impact on human life preservation. The current landscape of diagnostic methodologies for analyzing brain magnetic resonance imaging (MRI) scans shows potential for advancements in accuracy and timeliness [18, 28]. The academic discourse includes a range of papers, each exploring distinct methodologies for classifying brain tumors, highlighting the innovation in this sector [29–32]. The application of deep learning (DL) techniques in this context is noteworthy for their capacity to process and interpret complex medical imagery. These techniques have shown efficacy in handling diverse data types, such as high-dimensional images and multidisciplinary pathology scans [10, 33–36], establishing DL as a pivotal tool in the realm of biomedical image analysis and disease prognostication [11, 19, 37, 38].

The conventional approach of manually analyzing brain MRI data presents challenges in terms of both resource allocation and potential for human error, considering the specialized skill set required for accurate interpretation.

The proposition of an advanced, meticulously calibrated EfficientNet-based DL model specifically for brain tumor classification represents a significant leap forward in this field. Such a model is poised to substantially enhance the precision and efficiency of brain tumor diagnoses, thereby contributing to more effective and timely clinical decisions.

The manual analysis of brain MRI data is both a time-intensive and costly process, which is also susceptible to inaccuracies due to the specialized expertise required for processing and classifying these images. Accurate diagnosis and categorization of brain tumors are crucial, as they play a pivotal role in enabling medical professionals to make informed decisions and provide timely interventions.

To combat the high mortality rates associated with brain tumors, it is imperative to develop a robust deep learning (DL) model capable of swiftly and accurately predicting brain tumors. Our research is centered on devising an efficient and systematic framework for brain tumor classification. This involves several preprocessing steps to prepare our dataset, modify the transfer learning (TL) architecture, and integrate additional layers. We have tested our presented DL strategy on the publicly accessible Figshare MRI brain tumor dataset. Our novel DL methodology aims to enhance brain tumor classification significantly. The efficacy of our DL model will be evaluated using various metrics. Our DL model demonstrates the ability to classify brain tumors with a precision exceeding 99.50%.

The key contributions of this research are as follows:

(i) Development of an innovative EfficientNet-based DL model incorporating extensive preprocessing, TL architecture reconfiguration, and fine-tuning for effective brain tumor classification.

(ii) Introduction of image augmentation techniques into the architecture reconfiguration, aimed at enhancing model generalizability and robustness, while mitigating overfitting issues commonly encountered in deep learning models.

(iii) Integration of additional layers into the reconfiguration architecture, tailored to capture intricate features relevant to brain tumors, thereby enhancing the performance of the deep learning model on tumor classification.

The paper's organization is as follows: Section 2 presents an overview of related works in brain tumor classification using DL approaches. In Section 3, we describe in detail the methodology adopted in our research, along with the dataset employed. Section 4 focuses on the experimental setup, performance evaluation metrics, and description. The paper concludes with Section 6, summarizing the key findings and insights with future research.

2. Related Works

Talukder et al. [18] unveiled a novel DL strategy for the classification of brain tumors, encompassing stages such as data preprocessing, the establishment of a transfer learning framework, and meticulous fine-tuning. Their investigation evaluated various transfer learning models, including Xception, ResNet50V2, InceptionResNetV2, and DenseNet201, using a collection of 3,064 T1-weighted contrast-enhanced magnetic resonance imaging (CE-MRI) brain tumor images. The study's findings indicated exceptional accuracy rates, with ResNet50V2 emerging as the most effective model, achieving a 99.68% accuracy level. This performance surpassed that of competing models and holds the potential to enhance the speed and precision of brain tumor diagnoses in clinical settings.

Belaid et al. [29] introduced a DL methodology to classify three distinct types of brain tumors, meningioma, glioma, and pituitary tumors, employing the pretrained VGG-16 CNNs. The study not only utilized original images but also incorporated gray level cooccurrence matrix (GLCM) feature images as inputs to the CNN. The technique of using both the original and energy images as inputs significantly outperformed contemporary classifiers, achieving an average accuracy of 96.5% on T1-weighted CE-MRI dataset. The research's substantial contributions included the fusion of CNN and GLCM features to enhance accuracy, the adaptation of the VGG-16 model for training, and the extraction of GLCM feature images from the original images.

Huang et al. [30] developed CNNBCN, an innovative CNN architecture for brain tumor image classification within the medical imaging domain. This model demonstrated superior performance with a classification accuracy of 95.49%, exceeding many existing models. Additionally, it exhibited a lower test loss in comparison to well-known architectures such as ResNet, DenseNet, and MobileNet. Their work highlighted the advantages of CNNBCN in refining medical image analysis techniques. By integrating varied activation functions, they enhanced the model's accuracy by 0.5% to 0.75%, showcasing the CNNBCN model's effectiveness and its potential to rival or even surpass handcrafted models with reduced manual intervention.

Sejuti et al. [31] proposed a CNN-SVM-based approach aimed at improving the accuracy of brain tumor classification. The methodology began with constructing a 19-layer CNN featuring batch normalization and ReLU activation functions, which were applied to the T1-weighted CE-MRI dataset comprising 3,064 images across three tumor categories. Initial classification with Softmax achieved a high training accuracy of 96.74%, which was further elevated to 97.1% through the application of a multiclass support vector machine (SVM) that utilized features extracted by the CNN. This approach was distinguished by its minimal preprocessing of input data. Future research might explore alternative classifiers and preprocessing methods to evaluate their comparative effectiveness against the SVM.

Swati et al. [39] addressed the challenge of employing small datasets in DL for medical imaging, particularly in brain tumor classification. Their study utilized a pretrained deep CNN model and introduced a block-wise fine-tuning method based on transfer learning. Evaluating their approach on a T1-weighted CE-MRI benchmark dataset, they achieved an impressive average accuracy of 94.82% using five-fold cross-validation. This generic approach obviated the need for hand-crafted features and minimal preprocessing. Their comparative analysis underscored the method's effectiveness over traditional DL models applied to the CE-MRI dataset, indicating the transferability of insights from natural image processing to medical brain MR imaging.

Bashir et al. [40] presented the convolutional autoencoder neural network (CANN), which performed exceptionally well in brain tumor classification. The CANN achieved a 99.3% accuracy rate on the Cheng dataset, classifying tumors into meningioma, glioma, and pituitary categories. It also recorded a 99.1% accuracy in both glioma detection and grading on the IXI and BraTS 2017 datasets and a 98.5% success rate in sorting images into six distinct classes. The study's significant contributions included a one-step process for feature extraction and classification, leveraging both global and local features, training the network from scratch for architectural simplicity, and employing data augmentation to address data scarcity. By harnessing deep feature learning, the CANN outperformed other techniques, proving its reliability and potential in medical image classification.

Ayadi et al. [41] developed an innovative CNN-based model with multiple layers for classifying MRI brain tumors. This intelligent model required minimal preprocessing and was assessed using three distinct brain tumor datasets. A suite of performance metrics was employed to evaluate the model's accuracy and system robustness. The proposed model attained the highest classification accuracies in comparison to prior studies, with an accuracy rate of 94.74% on the Figshare dataset, 93.71% on the Radiopaedia dataset, and 97.22% on the Rembrandt dataset.

Paul et al. [42] utilized DL techniques to classify brain images into three tumor categories: meningioma, glioma, and pituitary tumors. They focused on 989 axial images from 191 patients within a dataset of 3,064 T1-weighted contrast-enhanced MRI brain images, to avoid confusion from different image planes. Their research compared two neural network architectures, fully connected networks and CNNs, and found that CNNs yielded superior performance, with an average accuracy of 93.00% in cross-validation. Their findings underscored the potential of DL in the medical domain for accurate brain tumor classification.

Alnowami et al. [43] focused on the development of an automated system for the detection of brain tumors in MRI scans using artificial neural networks. They worked with a dataset of 4,314 MRI images categorized into four groups: pituitary tumor, glioma, meningioma, and a normal healthy brain. Various preprocessing steps like intensity normalization and contrast enhancement were employed, and their impact on the accuracy of the model was assessed. A densely connected convolutional network (DenseNet) was trained on three different datasets. The study reported that preprocessing significantly enhanced tumor segmentation in automated systems employing DL techniques. The proposed model achieved an accuracy of 96.52%, with a sensitivity of 98.5% and specificity of 82.1%, when validated using ten-fold cross-validation. The research highlighted the importance of preprocessing steps in improving the effectiveness of tumor segmentation.

Arumugam et al. [44] presented an automated system designed to detect and classify brain tumors from MRI images accurately. The study introduced the Crossover Smell Agent Optimized Multilayer Perception (CSA-MLP) approach, aimed at precise identification and categorization of tumor cells. Utilizing three datasets of MRI images, preprocessing involved noise reduction and feature extraction. The classification phase relied on convolutional neural network (CNN) to distinguish between normal and tumorous brain cells, while the MLP was used to further classify tumor types, with enhancements from the CSA optimization algorithm leading to reduced errors and heightened performance. The experimental outcomes suggested that the proposed method surpasses existing techniques, achieving a remarkable accuracy rate of 98.56%, thus addressing the need for precise and automated brain tumor diagnostics using MRI.

Aamir et al. [45] developed a more efficient method for diagnosing brain cancers using MRI imaging. This approach improves image quality with a simple algorithm. A morphological analysis was conducted to remove nontumor regions before segmentation. Segmentation and clustering techniques identified high-quality tumor regions. Multiple deep neural networks extracted features from these ranked regions. An adaptive fusion network combined these features into a hybrid vector, which, along with a multiclass SVM, classified the tumors. To reduce overfitting, the training data were supplemented. The system was trained and tested on a T1-weighted CE-MRI dataset. Compared to previous methods, this approach enhanced the automation and robustness of the diagnostic process, achieving an accuracy of approximately 98.98%.

Aamir et al. [46] proposed an automated method for detecting brain tumors using MRI. First, brain MRI images were preprocessed to improve visual quality. Two different pretrained DL models were then used to extract features from the images. These feature vectors were combined into a hybrid vector using the partial least squares (PLS) method. Next, agglomerative clustering identified the top tumor locations. These locations were resized to a standard dimension and then passed to the head network for classification. This method achieved a classification accuracy of 98.95%, outperforming existing approaches.

Guan et al. [47] aimed to improve performance and reduce human effort with a new method for brain tumor detection in MRI images. First, the MRI images were preprocessed to enhance visual quality and increase the number of samples to prevent overfitting. Tumor locations were identified using an agglomerative clustering method. The images and these tumor locations were then sent to a backbone architecture for feature extraction. A refinement network further improved the quality of these tumor locations, discarding low-quality ones. The refined locations were resized to a uniform dimension and sent to a head network for classification. This method, tested on a publicly available brain tumor dataset, proved to be an effective tumor grading tool. Experimental results demonstrated that it outperformed existing methods, achieving a classification accuracy of 98.04%.

Qureshi et al. [48] introduced a novel MGMT Promoter Methylation Prediction (MGMT-PMP) system, combining radiomic and latent features for glioblastoma subtype prediction. A unique Deep Learning Radiomic Feature Extraction (DLRFE) module was proposed, integrating radiomic features with spatial distribution knowledge. The system employed a rejection algorithm to filter negative training instances. Fused feature vectors were used for training and testing with k-NN and SVM classifiers on the BraTS-2021 dataset. Through Jackknife tenfold cross-validation, the system achieved high classification performance, with accuracy, sensitivity, and specificity of $(96.84 \pm 0.09)\%$, $(96.08 \pm 0.10)\%$, and $(97.44 \pm 0.14)\%$, respectively, in detecting MGMT methylation status in glioblastoma patients. This approach, bridging genomic variation with radiomic features, holds promise for enhancing clinical treatment planning and improving patient outcomes.

3. Methodology

In our proposed approach, we aim to categorize brain tumors with high accuracy using EfficientNet models.

Figure 1 illustrates the proposed EfficientNet-based deep learning (DL) approach for brain tumor classification. The method depicted is a structured DL approach utilizing the EfficientNet architecture for classifying MRI brain tumor images.

The process begins with extensive preprocessing of the MRI dataset. Our preprocessing steps, which are crucial for standardizing input data, include resizing images to a standard size (224×224), converting them to RGB (if necessary), applying filtering techniques such as sharpening to enhance image quality, scaling pixel values for normalization, and accurately labeling the images. These steps ensure that the input data are consistent and clean, facilitating better performance of the model.

The processed images are then fed into a reconfigured EfficientNet algorithm (B0–B7). EfficientNet is selected for its efficient scaling and high performance in image classification tasks. We have incorporated data augmentation techniques such as horizontal flipping, random rotation, zooming, resizing, and translation to increase the diversity of the training dataset. This augmentation helps in reducing overfitting and improving the model's generalization capability.

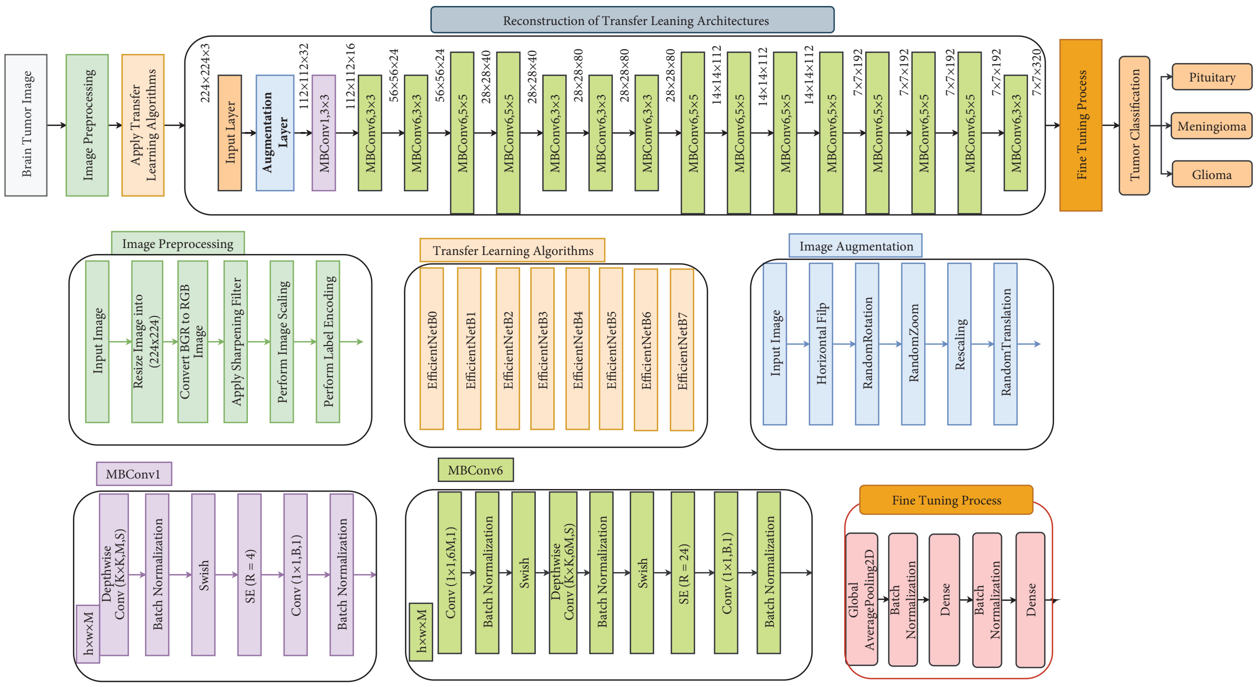


Figure 1: The proposed reconfigured and fine-tuned TL architecture for brain tumor classification.

Novel Contribution: One of the key contributions of our work is the integration and fine-tuning of pretrained EfficientNet models specifically for brain tumor classification. By leveraging transfer learning, we utilize the EfficientNet model+, pretrained on large image datasets and fine-tune them on our specific brain tumor dataset. This approach leads to enhanced performance and faster convergence compared to training from scratch.

Reconfiguration of EfficientNet: We have adapted the EfficientNet architecture by integrating additional layers tailored for brain tumor classification. These layers include modified MBConv blocks (MBConv1 and MBConv6) that are designed to better capture the intricate details of MRI images. The feature extraction layers are enhanced with techniques such as Swish activation, squeeze-and-excitation (SE) blocks, and batch normalization to improve the model's ability to learn from the complex data.

The network concludes with a classification layer that categorizes each image into one of three types of brain tumors: pituitary, meningioma, or glioma. The final classification layer uses a dense network followed by a Softmax activation to output the probability of each tumor type.

Hence, our contributions include a comprehensive preprocessing pipeline to standardize MRI images, ensuring consistent and clean input data. We utilize transfer learning with EfficientNet (B0–B7) models, which are fine-tuned specifically for brain tumor classification, leading to enhanced performance and faster convergence. To further improve model generalization, we incorporate data augmentation techniques. Additionally, we reconfigure the EfficientNet architecture by integrating additional layers tailored for improved feature extraction, enabling the model to better capture the intricate details of MRI images.

The transfer learning (TL) algorithms utilized in our methodology are due to leveraging pretrained models, such as EfficientNet, which have been trained on large-scale image datasets. By fine-tuning these pretrained models on our specific MRI brain tumor dataset, we can benefit from their learned features and hierarchical representations, thereby accelerating the training process and potentially improving the model's performance. Transfer learning allows us to transfer knowledge from tasks with abundant labeled data (in this case, generic image classification tasks) to tasks with limited labeled data (MRI brain tumor classification), ultimately aiding in the efficient training of our model and enhancing its ability to generalize to unseen data.

Our methodology diverges from existing deep learning (DL) techniques in several key ways:

(1) Innovative EfficientNet-based DL model: Unlike prior studies that primarily utilized conventional deep learning architectures (e.g., ResNet, DenseNet, and VGG), our research leverages the EfficientNet family, known for its superior performance and efficiency due to its compound scaling method. This allows our model to achieve higher accuracy with fewer computational resources.

(2) Extensive preprocessing and augmentation: Our approach incorporates a comprehensive preprocessing pipeline, including intensity normalization and contrast enhancement, which is not as extensively applied in previous works. Additionally, we employ sophisticated image augmentation techniques to address overfitting and enhance the robustness of our model.

(3) TL Architecture reconfiguration and fine-tuning: We reconfigure the transfer learning architecture specifically for brain tumor classification by adding and modifying layers within the EfficientNet model. This includes the introduction of additional layers that are tailored to capture the unique features of brain MRI images, optimizing the network's capacity to generalize from the training data.

(4) Immediate image standardization: We streamline the augmentation process by implementing immediate image standardization. This step ensures uniformity in the input data, reducing the variance and improving the model's ability to learn effectively from augmented images.

(5) Comprehensive evaluation metrics: Unlike some previous studies that focus on a limited set of performance metrics, we conduct an extensive evaluation using various metrics such as accuracy, precision, recall, F1-score, confusion matrix, sensitivity, specificity, AUC-ROC score, MAE, MSE, and RMSE. This thorough assessment allows us to identify the most effective model for brain tumor categorization with high reliability.

The primary contribution of our paper lies in demonstrating the effectiveness of the EfficientNet architecture in the domain of brain tumor classification, achieving state-of-the-art performance metrics and providing a detailed comparison with existing models. This research highlights the potential of our approach to enhance diagnostic accuracy and efficiency in clinical settings, paving the way for earlier and more precise medical interventions.

3.1. Dataset Collection. In our proposed model, we utilized a publicly available dataset [49] to assess the performance of our brain tumor classification networks. This comprehensive dataset consists of 2D MRI images, specifically contrast-enhanced T1-weighted magnetic resonance (CE-MRI) images, curated from a diverse cohort of 233 patients. The dataset encompasses images captured from three distinct anatomical views: coronal, sagittal, and axial views, providing a comprehensive representation of brain tumor data. The dataset comprises a total of 3064 MRI image slices, each meticulously annotated and categorized into three distinct brain tumor types. Specifically, it contains 708 slices corresponding to meningioma, 1426 slices representing glioma, and 930 slices depicting pituitary tumors. This diversity in tumor types and imaging views ensures the dataset's richness and suitability for training and evaluating our brain tumor classification models.

For reference, Figure 2 visually illustrates a representative sample from this dataset, offering a glimpse into the diverse and informative image data that our models were trained and tested on. Figure 3 shows the distribution of this dataset. This dataset serves as a valuable resource for advancing the field of medical image analysis and enhancing our understanding of brain tumor classification.

3.2. Image Preprocessing. In our brain tumor classification pipeline, comprehensive image preprocessing steps are applied to ensure that the input data are well prepared for the EfficientNet-based models. These preprocessing techniques enhance data quality, standardize image attributes, and facilitate effective model training.

(i) Image resizing (224×224).

To create uniformity in the dataset, all MRI images are resized to a standardized resolution of 224×224 pixels. This resizing step ensures consistent dimensions across the entire dataset, enabling efficient model training and feature extraction.

(ii) Channel reordering: To align with the input requirements of DL frameworks, a channel reordering step is performed. This step ensures that the grayscale MRI images have the correct channel order, aligning them with the expected RGB format for subsequent processing.

(iii) Image filtering for clarity: Image filtering techniques are applied to enhance the clarity and quality of the MRI images. These filters are carefully selected to reduce noise, improve contrast, and highlight relevant image features, making the images more suitable for accurate tumor classification.

(iv) Image scaling for numerical stability: Pixel values of the preprocessed images are scaled to a specific range, such as [0, 1] or [−1, 1]. This scaling step promotes numerical stability during model training, prevents convergence issues, and optimizes the learning process within the DL models.

(v) Image labeling: Each preprocessed MRI image is associated with a corresponding label indicating the tumor type it represents (meningioma, glioma, or pituitary tumor). This meticulous labeling process is crucial for supervised learning, enabling the models to accurately learn and generalize from the dataset.

These preprocessing steps collectively contribute to the refinement of our MRI image dataset, ensuring that it is clean, standardized, and well-suited for training and evaluating our brain tumor classification models. These optimized input data serve as the foundation for accurate and reliable brain tumor diagnosis.

Figure 4 visually demonstrates the impact of the preprocessing steps described in the text on the quality and clarity of the MRI images used in the brain tumor classification pipeline. The text accurately conveys this message by explaining that the preprocessing techniques enhance data quality and facilitate effective model training. The figure

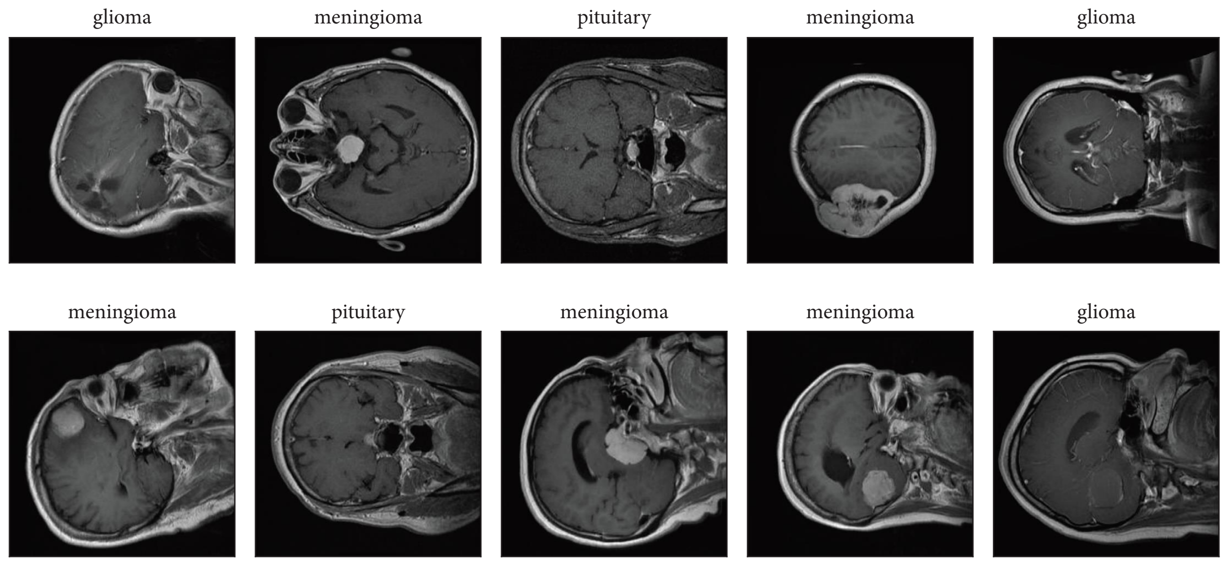


FIGURE 2: Sample brain tumor images.

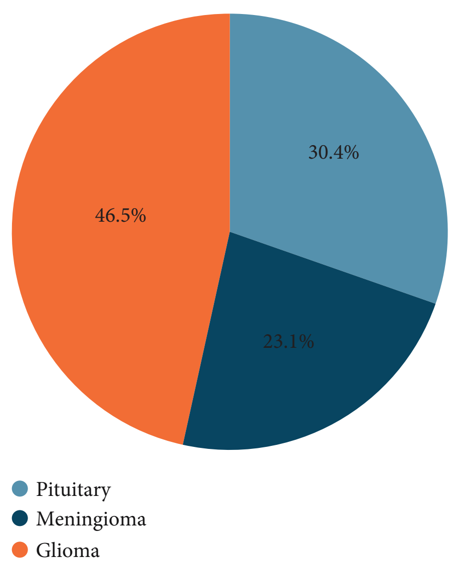


FIGURE 3: Distribution of the brain tumor dataset.

complements this explanation by providing visual evidence of how the preprocessing steps result in improved clarity, enhanced features, and greater visual distinction in the MRI images.

3.3. Reconfiguration of Transfer Learning Architectures. The reconfigured structure with fine-tuned EfficientNet (B0–B7) TL architecture is used for brain tumor classification. Fine-tuned EfficientNet (B0–B7) TL architecture refers to a series of deep learning architectures based on the EfficientNet family, including EfficientNet-B0 through EfficientNet-B7 which is EfficientNet-B0, EfficientNet-B1, EfficientNet-B2, EfficientNet-B3, EfficientNet-B4, EfficientNet-B5, EfficientNet-B6, and EfficientNet-B7 where the “Fine-tuned” aspect implies further optimizing the model’s layers during training to better suit our brain tumor classification. Therefore, it signifies a set of models derived from the EfficientNet family, which have been pretrained on general image data and fine-tuned for brain tumor classification. The architecture is visualized as a series of blocks, each representing different layers and operations within the neural network:

(i) Input layer: The beginning of the model where input data (images) are fed into the network.

(ii) Convolutional blocks (Block1, Block2, . . ., Block N): Each block in the network consists of a sequence of layers. These include

  (i) Convolutional (Conv2D) layers: Responsible for extracting features from the input images through filters.

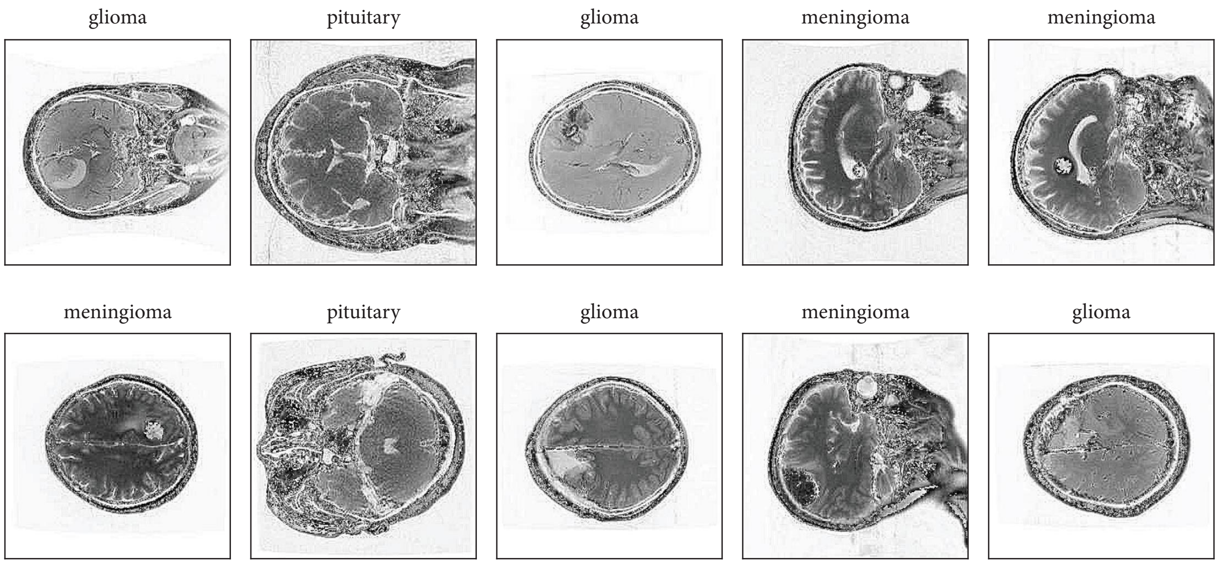


Figure 4: The preprocessed images of the brain tumor dataset.

  - (ii) Batch normalization: Normalizes the previous layer's output, reducing internal covariate shift and improving training stability.
  - (iii) Activation layers (like ReLU): Introduce nonlinearities into the model, allowing it to learn more complex patterns.
  - (iv) Pooling layers (like MaxPooling2D): Reduce the spatial dimensions of the output volume, decreasing the number of parameters and computations in the network.
  - (v) DepthwiseConv2D layers: Perform spatial convolutions over channels, optimizing the network's efficiency in processing multichannel data.

- (iii) Zero Padding2D layer: Adds zeros around the borders of the input matrix. This is useful for controlling the spatial dimensions of the output volumes, especially after convolutional operations.
- (iv) Global average Pooling2D layer: Compresses the feature map obtained from previous layers to a single vector per map by averaging the features. This significantly reduces the network's parameters and prepares it for the final classification stage.
- (v) Dense layer (fully connected layer): Comes after the feature extraction layers and is used for classification. These layers are densely connected, meaning each neuron receives input from all neurons of its preceding layer.
- (vi) Image augmentation layer: Presents the integration of data augmentation techniques such as random rotations, shifts, and zooms directly into the model. This is a strategic approach to increase the diversity of the training data, aiding the model in generalizing better and reducing the risk of overfitting.

  Activation layer (ReLU and Softmax): In this neural network architecture, the activation layer plays a pivotal role, employing two types of activation functions: ReLU (rectified linear unit) and Softmax.

  - (i) ReLU activation: Applied after convolutional or dense layers, ReLU introduces nonlinearities, allowing complex pattern learning. It outputs positives as-is and negatives as zero, enhancing efficiency and solving the vanishing gradient issue.
  - (ii) Softmax activation: Utilized in the final classification layer, Softmax converts logits into class probabilities, essential for multiclass tasks, and simplifies identifying the most probable class.

The reconfiguration process aims to exemplify how traditional EfficientNet architecture has been adapted or reconfigured with additional layers and operations to optimize it for the task of classifying brain tumors from MRI images. Each colored bar represents a different type of operation or layer within the network, and the sequence indicates the flow of data through the network from input to output.

3.3.1. Image Augmentation Layer. The "Image Augmentation Layer" refers to a series of transformations applied to input images before they are fed into a deep learning (DL) model. This layer is crucial in enhancing the robustness and generalization capabilities of the model. Below is an explanation of the transformations included in our image augmentation pipeline:

  - (i) Input image: This is the initial image that enters the augmentation pipeline. It serves as the base for all subsequent transformations.

(ii) RandomFlip("horizontal"): This transformation randomly flips the input image along the vertical axis (left to right). This helps the model learn to recognize objects regardless of their horizontal orientation.

(iii) RandomRotation(0.25): This transformation rotates the image randomly within a range of 25% of 360 degrees (i.e., up to 90 degrees in either direction). Such rotation increases variability and helps the model generalize better to different orientations.

(iv) RandomZoom(0.25): This transformation randomly zooms in or out of the image by 25%. This variation in scale helps the model recognize objects of different sizes.

(v) RandomContrast(0.25): This transformation adjusts the image contrast by up to 25%, enhancing different features within the images and improving the model's ability to detect subtle differences.

(vi) RandomRotation(35): This transformation applies an additional rotation of 35 degrees to further introduce rotational variance, although this might be a fixed value rather than a random one. This enhances the model's ability to handle different image orientations.

(vii) Incorporating an image augmentation layer as part of the data preprocessing pipeline serves several important purposes:

(viii) Increase dataset diversity: By creating multiple variants of the original images, augmentation effectively enlarges the training dataset, helping the model see more diverse examples without additional data collection.

(ix) Prevent overfitting: Augmentation introduces variations that help the model learn general patterns rather than memorizing specific details of the training data, thus improving performance on unseen data.

(x) Improve generalization: Augmented images simulate a wide array of real-world conditions, ensuring the model performs well across different scenarios.

By utilizing these augmentation techniques, we enhance the model's robustness and ensure it can maintain high performance across diverse and unseen datasets, ultimately leading to better and more reliable brain tumor classification.

3.3.2. Model Bias and Mitigating Biases. To address model bias and mitigate biases stemming from the dataset's lack of representation across diverse demographics or tumor types and to strengthen the model's applicability, we have implemented several strategies:

(1) Data augmentation: To mitigate bias stemming from imbalanced datasets, we employ data augmentation techniques. By artificially increasing the diversity of the training data through techniques such as rotation, flipping, and scaling, we ensure that the model is exposed to a broader range of features and scenarios.

(2) Transfer learning adaptation: Our model leverages transfer learning from pretrained EfficientNet architectures. This approach allows the model to inherit knowledge from a large dataset used for pretraining, which typically includes diverse samples. By fine-tuning the model on our specific dataset, we adapt it to the nuances of brain tumor classification while retaining the generalizability acquired during pretraining.

(3) Bias detection and mitigation: We continuously monitor the model's performance when training the model across different demographic groups or tumor types to detect and address potential biases. This involves analyzing performance metrics disaggregated by demographic variables and tumor categories. If biases are detected, we implement corrective measures such as model adjustments with learning rates to mitigate them.

By implementing these strategies, we aim to minimize model bias and enhance the applicability and fairness of our proposed model in real-world clinical settings.

3.4. EfficientNet Models. EfficientNet is a family of convolutional neural network (CNN) architectures that have set new benchmarks in terms of efficiency and accuracy for image classification tasks. These models were developed using a method known as compound scaling, which uniformly scales network width, depth, and resolution with a fixed set of scaling coefficients. This approach helps achieve better performance with fewer resources compared to traditional scaling methods.

EfficientNet models range from B0 to B7, each progressively larger and more powerful. Below, we provide an overview of each model and its characteristics:

(i) EfficientNetB0: The baseline model in the EfficientNet family. It has a moderate number of parameters and computational cost, making it a good starting point for various image classification tasks. EfficientNetB0 balances efficiency and accuracy effectively.

(ii) EfficientNetB1: Builds on B0 by increasing the network's depth and width, leading to improved accuracy. This model maintains computational efficiency while offering a modest performance boost over B0.

(iii) EfficientNetB2: Further scales up the depth and width compared to B1. EfficientNetB2 offers enhanced accuracy and is suitable for more demanding tasks, while still being computationally efficient.

(iv) EfficientNetB3: Introduces significant increases in both depth and width over B2, providing higher accuracy for complex image classification tasks. This model is particularly effective when high accuracy is required without a substantial increase in computational cost.

(v) EfficientNetB4: Expands further on B3, featuring an even deeper and wider architecture. EfficientNetB4 is designed for high-accuracy applications, making it ideal for challenging tasks like medical image classification.

(vi) EfficientNetB5: Continues the trend of increasing depth and width, offering greater accuracy and capacity for more complex tasks. This model is well suited for scenarios requiring fine-grained classification and high detail recognition.

(vii) EfficientNetB6: Extends the architecture further, making it capable of handling very demanding image classification tasks with high accuracy. EfficientNetB6 balances increased computational requirements with significant performance improvements.

(viii) EfficientNetB7: The largest and most powerful model in the EfficientNet family. It offers state-of-the-art performance for the most challenging image classification tasks, including large-scale and high-resolution image datasets.

The choice of EfficientNet models for our brain tumor classification task is driven by several key advantages:

(i) Efficiency and scalability: EfficientNet models are designed to maximize accuracy while minimizing computational resources. This makes them ideal for medical imaging applications where computational efficiency is critical.

(ii) Advanced feature extraction: Pretrained on large and diverse image datasets, EfficientNet models can effectively extract complex features from medical images, enhancing their ability to recognize and classify subtle patterns in brain MRI scans.

(iii) Proven performance: EfficientNet models have demonstrated superior performance in numerous benchmarks and real-world applications, providing confidence in their ability to achieve high accuracy in brain tumor classification tasks.

By leveraging the strengths of EfficientNet models, we aim to develop a highly accurate and efficient system for brain tumor classification, ultimately improving patient diagnosis and treatment.

3.5. Adaptive Fine-Tuning Process. The adaptive fine-tuning process is a critical step in optimizing the performance of our deep learning model for the specific task of brain tumor classification. This process involves several stages each designed to enhance the model's ability to learn from the training data and generalize to unseen data.

(i) Initialization: We start by setting a random seed to ensure the reproducibility of our results. The number of classes is set to three, corresponding to the different types of brain tumors (pituitary, meningioma, and glioma).

(ii) Input layer: The input layer is designed to accept images of a fixed size (e.g., 224×224 pixels) with three channels (RGB). This standardizes the input format and prepares the images for further processing.

(iii) Data augmentation: To enhance the robustness of the model and prevent overfitting, we apply a series of data augmentation techniques. These include

   (i) Random horizontal flip: Randomly flips the images horizontally, increasing variability.

   (ii) Random rotation: Rotates the images by a random angle within a specified range, simulating different viewing angles.

   (iii) Random zoom: Zooms in and out of the images randomly to simulate different scales.

   (iv) Random contrast: Adjusts the contrast of the images randomly to simulate different lighting conditions.

(iv) Preprocessing: After augmentation, the images are preprocessed to normalize pixel values, ensuring that they fall within a range suitable for the neural network.

(v) Base model: A pretrained EfficientNet model (e.g., EfficientNetB3) serves as the base model. This model has been trained on a large dataset and can extract powerful features from the images.

(vi) Global average pooling (GAP): The output from the base model is passed through a GAP layer, which reduces the spatial dimensions to a single vector. This step reduces the number of parameters and helps prevent overfitting.

(vii) Dense layers: A dense layer with 1280 units and ReLU activation is added to learn complex representations. Batch normalization is applied to stabilize and speed up training.

(viii) Output layer: The final dense layer uses a Softmax activation function to output the probability distribution over the three classes.

The entire model, including the pretrained base and newly added layers, is set to be trainable. This allows for fine-tuning of the weights to better adapt to the specific features of brain tumor MRI images. The model is compiled with the Adam optimizer, using a learning rate of 0.0001. The loss function used is sparse categorical cross-entropy, and accuracy is tracked as the primary metric.

The fine-tuning block is crucial for adapting the pretrained EfficientNet model to the specific task of brain tumor classification. By leveraging a model pretrained on the extensive ImageNet dataset, we benefit from its learned general features. Transfer learning allows us to fine-tune these features for our specialized dataset, focusing on brain MRI images. We freeze the initial layers, which capture generic features and fine-tune the later layers to learn task-specific features. Data augmentation techniques are applied to enhance robustness and generalizability, simulating various real-world imaging conditions. Global average pooling reduces spatial dimensions, and dense layers with ReLU activation functions and batch normalization help the model learn complex representations. The Softmax activation in the final layer provides probabilistic outputs for classification. Using the Adam optimizer and sparse categorical cross-entropy loss function ensures effective training and accurate classification. This approach ensures the model is well adapted to the specific characteristics of brain tumor images, improving its accuracy and reliability. Overall, the combination of EfficientNet models and an adaptive fine-tuning process ensures that our system can accurately classify brain tumors, providing valuable support for medical diagnosis and treatment.

4. Result Analysis

In this section, we delve into the in-depth performance analysis of several EfficientNet-based DL algorithms employed in the classification of brain tumors. Through rigorous evaluation and assessment, we aim to uncover the strengths and weaknesses of these models, ultimately identifying the most effective approach for accurate brain tumor classification.

In our proposed approach, we aim to categorize brain tumors with high accuracy using EfficientNet models. Figure 1 illustrates the proposed EfficientNet-based deep learning (DL) approach for brain tumor classification. The method depicted is a structured DL approach utilizing the EfficientNet architecture for classifying MRI brain tumor images.

The process begins with extensive preprocessing of the MRI dataset, including resizing images to a standard size, converting them into grayscale, applying filtering techniques to remove noise, scaling pixel values, and accurately labeling the images. This preprocessing ensures that the input data are standardized, facilitating better performance of the model. The processed images are then fed into a reconfigured EfficientNet model, which is known for its efficient scaling and high performance in image classification tasks. We have adapted the EfficientNet architecture by integrating additional layers specifically tailored for brain tumor classification. This includes layers for feature extraction, which enhance the model's ability to capture intricate details of the MRI images. The network concludes with a classification layer that categorizes each image into one of three types of brain tumors: pituitary, meningioma, or glioma. By leveraging the scalable and efficient architecture of EfficientNet, our method achieves accurate and reliable classification of medical images.

To ensure the robustness of our approach, we incorporate techniques such as data augmentation and transfer learning. Data augmentation helps in increasing the diversity of the training dataset, thereby reducing overfitting and improving the model's generalization capability. Transfer learning allows us to utilize pretrained EfficientNet models, which are fine-tuned on our specific dataset, leading to enhanced performance and faster convergence.

In summary, our proposed methodology combines comprehensive preprocessing, advanced EfficientNet architecture, and effective training techniques to achieve state-of-the-art performance in brain tumor classification, ultimately aiding in accurate and timely medical diagnoses.

4.1. Experiment Setup. This study was conducted on a robust computing setup equipped with an Intel Xeon CPU encompassing 3 cores and 15 GB of RAM, complemented by a 26 GB GPU, ensuring efficient processing capabilities. The system's 50 GB hard drive provided ample storage for the extensive data and models used in the research. The experimental framework was established within a Jupyter Notebook environment, which offered a flexible and interactive platform for conducting the analyses. The implementation of our proposed methodology was carried out using Python, a versatile programming language that offers comprehensive support for data analysis and machine learning tasks. To augment Python's core functionality, a suite of well-established libraries was employed. These included Scikit-learn for machine learning algorithms, Keras and TensorFlow for deep learning model development, Seaborn and Matplotlib for data visualization, and Numpy and Pandas for data manipulation and analysis.

Our experimental design incorporated the use of k-fold cross-validation to ensure the robustness and generalizability of the model. In this approach, the dataset was divided into ten distinct subsets ($k = 10$), allowing for thorough evaluation. The distribution of data was carefully managed, with 70% allocated for training the models, 20% used for validation purposes to fine-tune the model parameters, and the remaining 10% reserved for testing and assessing the model's performance. This structured approach to data management and model evaluation was critical in achieving the high levels of accuracy reported in our findings.

4.2. Performance Metrics. The key performance metrics used in our experiment, including accuracy, precision, recall, F1-score, mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) [50], are as follows:

(i) Accuracy: Accuracy is mostly used for classification problems.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \quad (1)$$

(ii) Precision: Precision quantifies the accuracy of positive predictions by calculating the ratio of correctly predicted positive observations to the total predicted positive observations.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive + False Positive}}. \tag{2}$$

(iii) Recall (also known as sensitivity or true positive rate): *Recall* assesses the ability of a model to capture all positive instances by determining the ratio of correctly predicted positive observations to all the observations in the actual class.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive + False Negatives}}. \tag{3}$$

(iv) F1-Score: The *F1-Score* gauges the balance between Precision and Recall, offering a weighted average that considers both false positives and false negatives.

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision + Recall}}. \tag{4}$$

(v) Sensitivity: The proportion of actual positives correctly identified by the model.

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive + False Negative}}. \tag{5}$$

(vi) Specificity: The proportion of actual negatives correctly identified by the model.

$$\text{Specificity} = \frac{(\text{True Negatives})}{(\text{True Negatives + False Positive})}. \tag{6}$$

(vii) AUC-ROC: The AUC-ROC score is calculated as the area under the ROC curve, which plots the true positive rate (sensitivity) against the false positive rate (1 - specificity) at various threshold settings. AUC-ROC ranges from 0 to 1, where 1 indicates a perfect model and 0.5 indicates a model with no discrimination ability.

(viii) Mean absolute error (MAE): *MAE* is the average of the absolute errors, where y_i is the actual value and $\hat{y}_i$ is the predicted value.

$$\text{MAE} = \frac{1}{N} \sum_{i=0}^{n} y_i - \hat{y}_i. \tag{7}$$

(ix) Mean squared error (MSE): *MSE* is the average of the squares of the errors. The larger errors are more heavily penalized than smaller ones, making MSE more sensitive to outliers than MAE.

$$\text{MSE} = \frac{1}{N} \sum_{i=0}^{n} \left(y_i - \hat{y}_i\right)^2. \tag{8}$$

(x) Root mean squared error (RMSE): *RMSE* is the square root of the mean squared error. It is more sensitive to outliers than MAE and is often used in regression analysis.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=0}^{n} \left(y_i - \hat{y}_i\right)^2}. \tag{9}$$

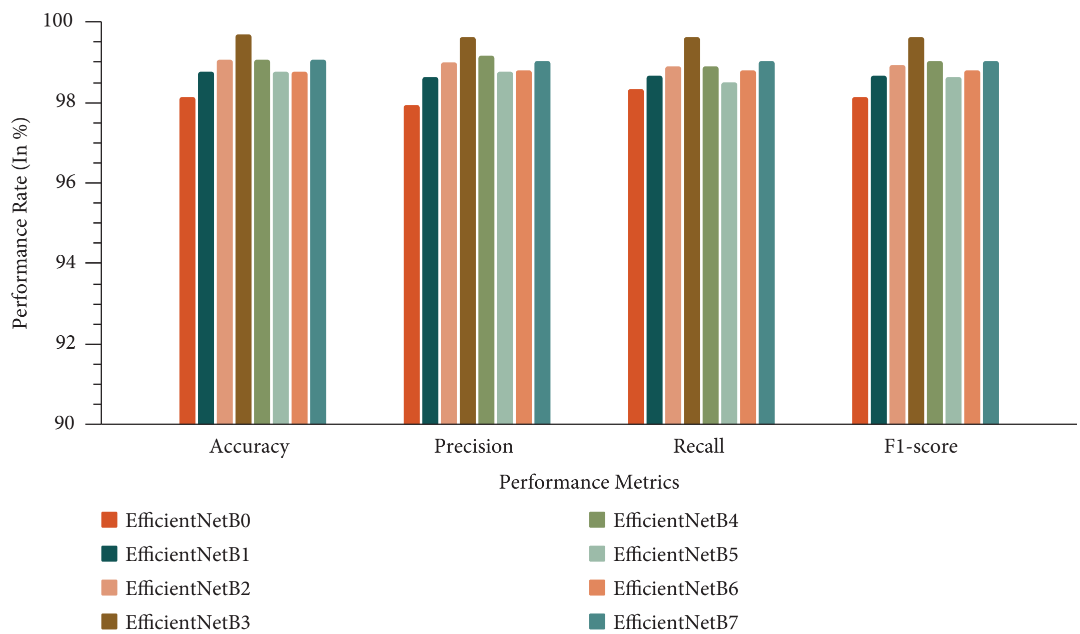
4.3. Performance Analysis. The performance analysis of different EfficientNet models for brain tumor classification is summarized in Table 1.

This table provides a detailed breakdown of key performance metrics, including accuracy, precision, recall, F1-score, MAE, MSE, and RMSE. To provide a visual representation of the performance, Figure 5 displays corresponding graphs. These metrics offer insights into the models' effectiveness in classifying glioma, meningioma, and pituitary brain tumors.

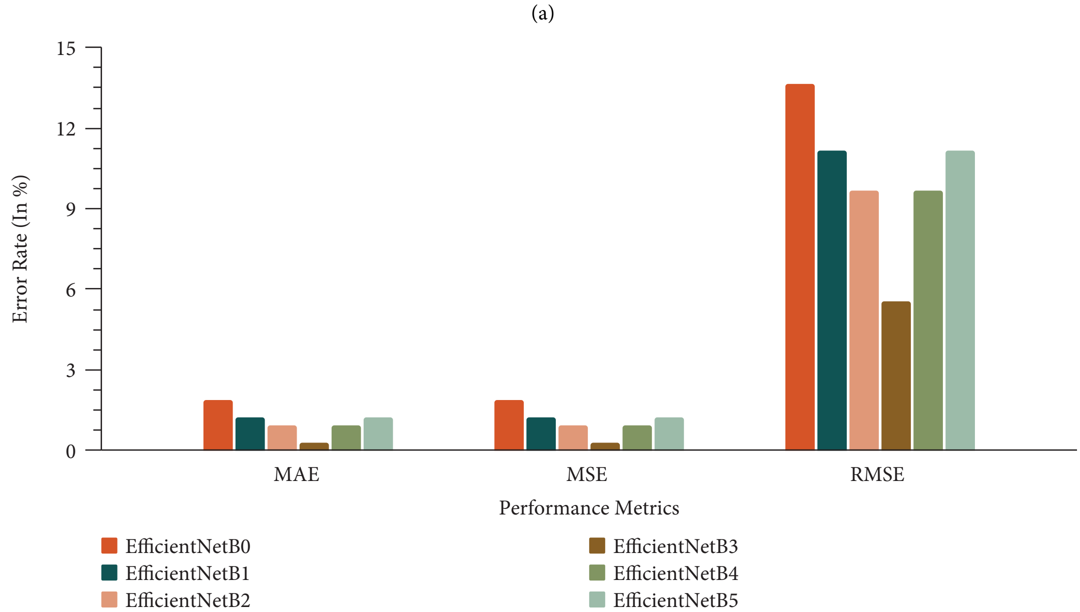
The results reveal that the EfficientNetB0 model achieved an accuracy of 98.14%, a precision of 97.95%, a recall of 98.35%, and an F1-score of 98.14%. Additionally, it displayed a sensitivity of 98.35%, a specificity of 99.09%, and an AUC-ROC score of 98.72%. The error metrics for this model included an MAE of 1.86, MSE of 1.86, and RMSE of 13.65. EfficientNetB1 exhibited even higher performance, with an accuracy of 98.76%, a precision of 98.64%, a recall of 98.67%, and an F1-score of 98.65. The sensitivity and specificity were 98.67% and 99.38%, respectively, and the AUC-ROC score was 99.03. The MAE, MSE, and RMSE for this model were 1.24, 1.24, and 11.15, respectively. EfficientNetB2 demonstrated superior performance with an accuracy of 99.07%, a precision of 99.01%, a recall of 98.90%, and an F1-score of 98.94. Its sensitivity was 98.90%, specificity was 99.53%, and AUC-ROC score was 99.22. The error metrics were impressively low, with an MAE of 0.93, MSE of 0.93, and RMSE of 9.65. The EfficientNetB3 model achieved an accuracy of 99.69%, a precision of 99.62%, a recall of 99.63%, and an F1-score of 99.62%. Additionally, it displayed a sensitivity of 99.63%, a specificity of 99.86%, and an AUC-ROC score of 99.75. The error metrics included an MAE of 0.31, MSE of 0.31, and RMSE of 5.57. The EfficientNetB4 model achieved an accuracy of 99.07%, a precision of 99.16%, a recall of 98.88%, and an F1-score of 99.02%. Additionally, it displayed a sensitivity of 98.88%, a specificity of 99.48%, and an AUC-ROC score of 99.18. The error metrics included an MAE of 0.93, MSE of 0.93, and RMSE of 9.65. The EfficientNetB5 model achieved an accuracy of 98.76%, a precision of 98.78%, a recall of 98.51%, and an F1-score of 98.64%. Additionally, it displayed a sensitivity of 98.51%, a specificity of 99.34%, and an AUC-ROC score of 98.93. The error metrics included an MAE of 1.24, MSE of 1.24, and RMSE of 11.15. The EfficientNetB6 model achieved an accuracy of 98.76%, a precision of 98.81%, a recall of 98.81%, and an F1-score of 98.81%. Additionally, it displayed a sensitivity of 98.81%, a specificity of 99.33%, and an AUC-ROC score of 99.07. The error metrics included an MAE of 1.24, MSE of 1.24, and RMSE of 11.15. Finally, the EfficientNetB7 model achieved an accuracy of 99.07%,

Table 1: Performance analysis of various EfficientNet models.

EfficientNet model	Accuracy	Precision	Recall	F1-score	Sensitivity	Specificity	AUC-ROC	MAE	MSE	RMSE
EfficientNetB0	98.14	97.95	98.35	98.14	98.35	99.09	98.72	1.86	1.86	13.65
EfficientNetB1	98.76	98.64	98.67	98.65	98.67	99.38	99.03	1.24	1.24	11.15
EfficientNetB2	99.07	99.01	98.9	98.94	98.9	99.53	99.22	0.93	0.93	9.65
EfficientNetB3	99.69	99.62	99.63	99.62	99.63	99.86	99.75	0.31	0.31	5.57
EfficientNetB4	99.07	99.16	98.88	99.02	98.88	99.48	99.18	0.93	0.93	9.65
EfficientNetB5	98.76	98.78	98.51	98.64	98.51	99.34	98.93	1.24	1.24	11.15
EfficientNetB6	98.76	98.81	98.81	98.81	98.81	99.33	99.07	1.24	1.24	11.15
EfficientNetB7	99.07	99.02	99.04	99.02	99.04	99.53	99.28	0.93	0.93	9.65



(a)



(b)

Figure 5: The performance results of the brain tumor dataset. (a) Performance. (b) Error.

a precision of 99.02%, a recall of 99.04%, and an F1-score of 99.02%. Additionally, it displayed a sensitivity of 99.04%, a specificity of 99.53%, and an AUC-ROC score of 99.28. The error metrics included an MAE of 0.93, MSE of 0.93, and RMSE of 9.65.

The table shows that the EfficientNetB3 algorithm achieves the highest accuracy rate of 99.69% among the analyzed algorithms. Furthermore, EfficientNetB3 performs better than other algorithms in terms of F1-score, recall, and precision. These findings highlight EfficientNetB3's superior performance in accurately classifying brain tumor cases. Additionally, the EfficientNetB3 model exhibits outstanding performance in terms of sensitivity, specificity, and ROC-AUC score. Specifically, it has a sensitivity of 99.63%, a specificity of 99.86%, and an ROC-

AUC score of 99.75, indicating its excellent ability to correctly identify positive cases and accurately distinguish between classes. The EfficientNetB3 algorithm also demonstrates exceptional performance in terms of error metrics, with low MAE, MSE, and RMSE values. These metrics measure the discrepancy between expected and actual values, with the EfficientNetB3 model achieving an MAE of 0.31, MSE of 0.31, and RMSE of 5.57, which further emphasizes the model's accuracy and reliability. These metrics measure the discrepancy between expected and actual values, and the model's dependability in producing precise predictions is demonstrated by the consistently low error rates. Overall, the EfficientNetB3 algorithm shows exceptional performance across all evaluation metrics, including sensitivity, specificity, and ROC-AUC score, achieving the highest accuracy rate and making it a promising option for brain tumor detection.

The results for the other EfficientNet models (B3, B4, B5, B6, and B7) also reflect high accuracy and precision scores, making them promising choices for brain tumor classification tasks. These models offer a range of options with varying trade-offs in error metrics.

4.4. Accuracy Graph Analysis. Figure 6 shows a series of accuracy graphs for various configurations of the EfficientNet DL model, specifically tailored for the task of brain tumor classification. These graphs display the performance of different versions of the EfficientNet model, from B0 to B7, over a series of epochs during the training process.

Each subgraph, labeled from (a) to (h), corresponds to a different version of the EfficientNet architecture (B0 to B7, respectively). Two lines are plotted in each graph: one for training accuracy and the other for validation accuracy. The x-axis represents the number of epochs, which is a full iteration over the entire dataset used for training, and the y-axis represents the accuracy, with a scale from about 0.65 to 1.00, where 1.00 denotes 100% accuracy.

From the provided title and data, we can deduce that the models have been optimized and fine-tuned for the specific task of brain tumor classification, and the accuracy rates for each model variant are given. These rates are very high, ranging from 98.14% to 99.69%, indicating that the models are highly effective at this task.

The accuracy values mentioned in your text seem to correspond to the validation accuracy at the end of the training period, i.e., after the final epoch. For instance, EfficientNetB3 achieves the highest validation accuracy of 99.69%, which aligns with the highest peak in the corresponding graph (d). The graphs demonstrate the models' learning progress, with training typically starting with lower accuracy and improving rapidly over a few epochs until it plateaus or improves more gradually. The validation accuracy also improves, but it can sometimes plateau or slightly decrease, which could be indicative of overfitting if the training accuracy continues to increase while validation accuracy does not. These graphs are an important tool for DL practitioners to evaluate and compare the performance of different model architectures or configurations and to monitor for overfitting, where the model learns the training data too well and fails to generalize to new, unseen data.

4.5. Loss Graph Analysis. Figure 7 displays a set of loss graphs for the EfficientNet DL models, ranging from B0 to B7. These graphs are typically used alongside accuracy graphs to provide a fuller picture of a model's learning process over the course of training.

Each graph, labeled from (a) to (h), corresponds to a specific variant of the EfficientNet model. The x-axis of each graph represents the epoch number, which is indicative of the number of complete passes the algorithm has made through the entire training dataset. The y-axis represents the loss value, which quantifies the error between the predicted values and the actual values, with lower values indicating better performance.

Two lines are plotted in each graph: *Training Loss*: This line represents the loss on the training dataset as the model learns. Ideally, this should decrease steadily as the number of epochs increases, indicating that the model is learning to fit the training data better over time. *Validation Loss*: This line represents the loss on a separate validation dataset that is not used for training. This metric helps measure how well the model is generalizing to new, unseen data.

For all models (EfficientNetB0 to EfficientNetB7), both training and validation losses decrease rapidly in the first few epochs, which suggests that the models are learning effectively. A key observation is whether the validation loss plateaus or starts to increase after decreasing, which could be a sign of overfitting, however, that does not appear to be the case here as the validation loss seems to stabilize or continue to decrease slightly, which is indicative of good generalization.

Among the EfficientNet models depicted in the loss graphs, EfficientNetB3 demonstrates the most promising performance with the lowest validation loss, which suggests that it has achieved the best generalization on the unseen validation data. This indicates EfficientNetB3 may be the most effective model for the task of brain tumor classification, balancing well between learning the training data and generalizing to new data. Its loss curve shows a rapid initial decrease and then stabilizes, which is characteristic of a well-fitting model.

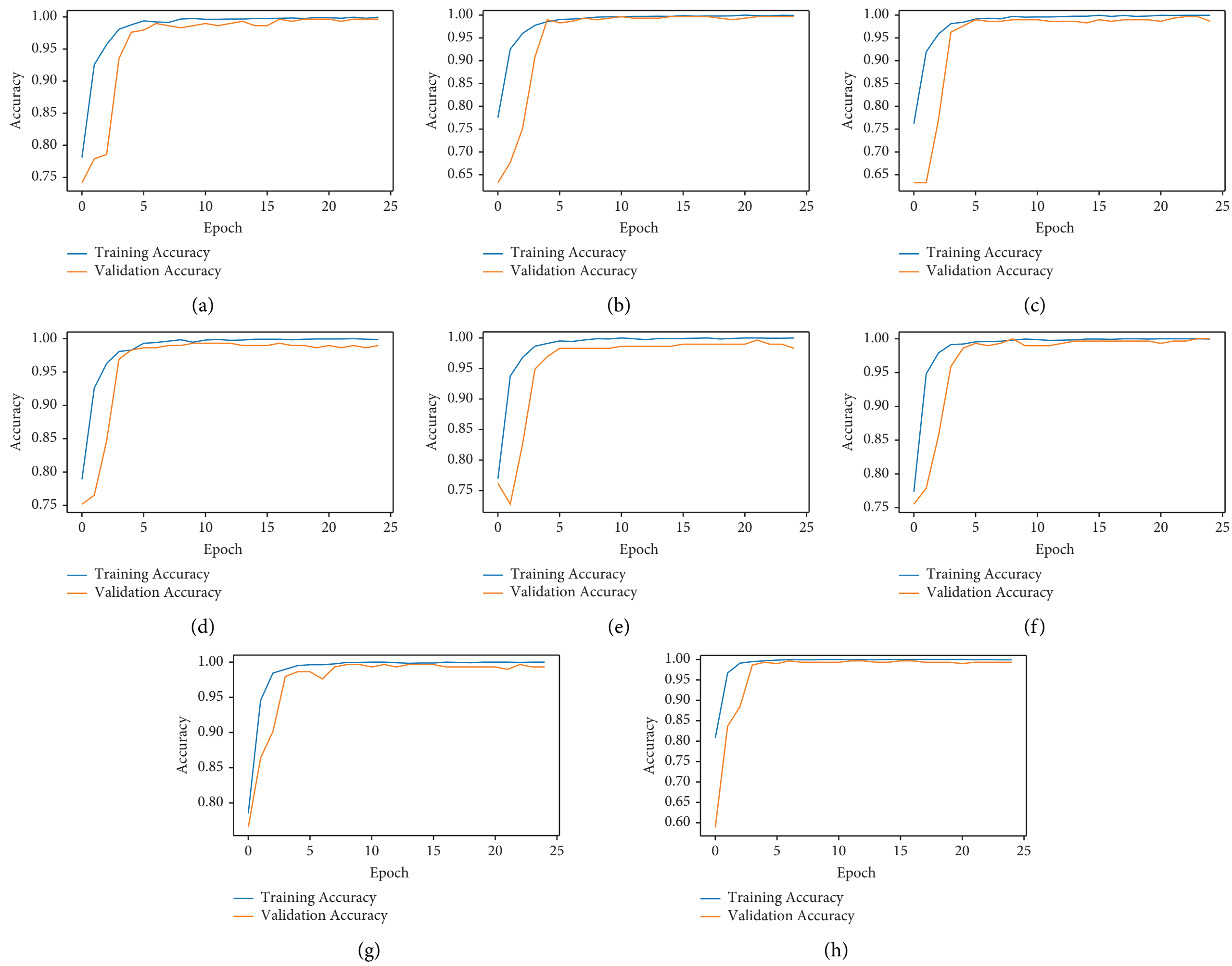


Figure 6: Accuracy graph of EfficientNet models. (a) EfficientNetB0. (b) EfficientNetB1. (c) EfficientNetB2. (d) EfficientNetB3. (e) EfficientNetB4. (f) EfficientNetB5. (g) EfficientNetB6. (h) EfficientNetB7.

4.6. Confusion Matrix Analysis. The confusion matrix provides a breakdown of the predicted and actual positive/negative classes, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Figure 8 displays the confusion matrices for various EfficientNet models (B0 to B7) used for classifying brain tumors into three classes: glioma, meningioma, and pituitary tumors. A confusion matrix is a table used to describe the performance of a classification model on a set of test data for which the true values are known. Each confusion matrix shows the number of true positive predictions along the diagonal (top-left to bottom-right) for each class, which indicates the number of times the model correctly predicted the class of the tumor. The off-diagonal numbers represent misclassifications, where the model predicted a different class than the actual class.

For example, the EfficientNetB0 model's confusion matrix shows that for glioma 141 TP, 176 TN, 1 FP, 4 FN; for meningioma 89 TP, 227 TN, 4 FP, 2 FN; for pituitary 86 TP, 235 TN, 1 FP, 0 FN; the EfficientNetB1 model's confusion matrix shows that for glioma 144 TP, 176 TN, 1 FP, 1 FN; for meningioma 89 TP, 230 TN, 1 FP, 3 FN; for pituitary 86 TP, 234 TN, 2 FP, 0 FN; the EfficientNetB2 model's confusion matrix shows that for glioma 145 TP, 176 TN, 1 FP, 0 FN; for meningioma 88 TP, 231 TN, 0 FP, 3 FN; for pituitary 86 TP, 234 TN, 2 FP, 0 FN; the EfficientNetB3 model's confusion matrix shows that for glioma 145 TP, 177 TN, 0 FP, 0 FN; for meningioma 90 TP, 231 TN, 0 FP, 1 FN; for pituitary 86 TP, 235 TN, 1 FP, 0 FN; the EfficientNetB4 model's confusion matrix shows that for glioma 145 TP, 175 TN, 2 FP, 0 FN; for meningioma 88 TP, 231 TN, 0 FP, 3 FN; for pituitary 86 TP, 235 TN, 1 FP, 0 FN; the EfficientNetB5 model's confusion matrix shows that for glioma 145 TP, 175 TN, 2 FP, 0 FN; for meningioma 88 TP, 230 TN, 1 FP, 3 FN; for pituitary 85 TP, 235 TN, 1 FP, 1 FN; the EfficientNetB6 model's confusion matrix shows that for glioma 143 TP, 175 TN, 2 FP, 2 FN; for meningioma 89 TP, 229 TN, 2 FP, 2 FN; for pituitary 86 TP, 236 TN, 0 FP, 0 FN; the EfficientNetB7 model's confusion matrix shows that for glioma 44.72 TP, 54.66 TN, 0.31 FP, 0.31 FN; for meningioma 27.64 TP, 71.43 TN, 0.31 FP, 0.62 FN; for pituitary

26.71 TP, 72.98 TN, 0.31 FP, 0 FN; to determine the best model, we would look for the one with the highest true positive rates and the lowest false positives and false

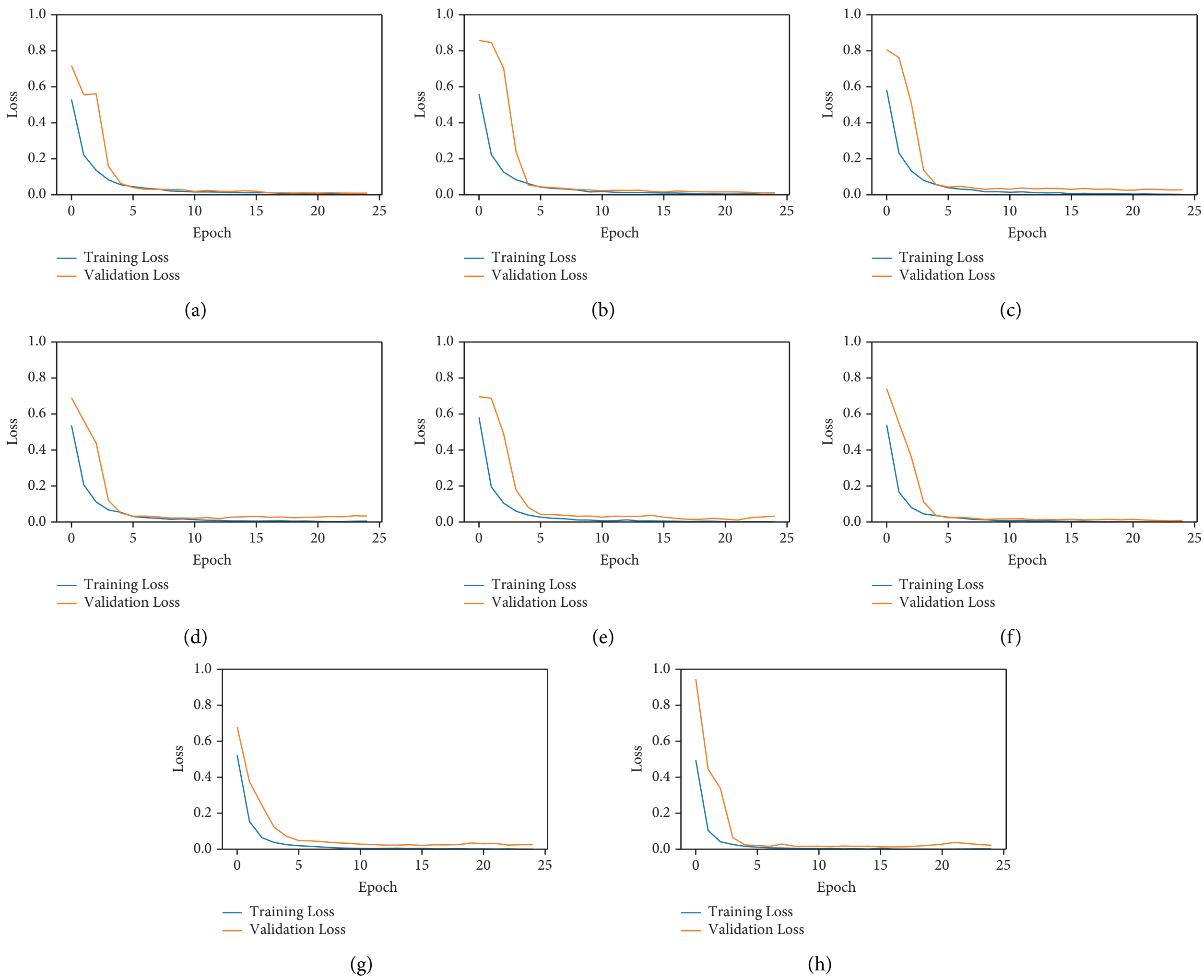


Figure 7: Loss graph of EfficientNet models. (a) EfficientNetB0. (b) EfficientNetB1. (c) EfficientNetB2. (d) EfficientNetB3. (e) EfficientNetB4. (f) EfficientNetB5. (g) EfficientNetB6. (h) EfficientNetB7.

negatives. Without numerical accuracy data, it is not possible to declare the "best" model purely based on these matrices, but visually, models like EfficientNetB3 and EfficientNetB7, which have the least number of misclassifications, could be considered among the top performers. The models' superior performance can be attributed to their architectural nuances, which make them well suited for this specific classification task: glioma, meningioma, and pituitary.

From the performance results of various matrices, it is evident that the EfficientNetB3 model stands out as the top-performing model with the highest accuracy of 99.69%. This indicates that EfficientNetB3 achieved the best overall classification performance for breast cancer detection among the evaluated models.

The superior performance of EfficientNetB3 can be attributed to several factors:

(i) Architecture complexity: EfficientNetB3, being a deeper and more complex architecture compared to some other variants, can capture more intricate patterns and features in breast cancer images. This increased depth allows it to learn a broader range of distinguishing characteristics.

(ii) Capacity for representation: EfficientNetB3 has a larger capacity for representing image features, which enables it to effectively differentiate between benign and malignant tumors. This capacity may have led to a more robust and accurate model.

(iii) Generalization: EfficientNetB3's architecture appears to generalize well to the breast cancer detection task, as indicated by its high accuracy. It strikes a balance between complexity and generalization, making it an optimal choice for this specific classification problem.

(iv) Fine-tuning: The fine-tuning process employed in EfficientNetB3 might have been particularly effective in adapting the model to the nuances of breast cancer classification. Fine-tuning allows the model to specialize in this task.

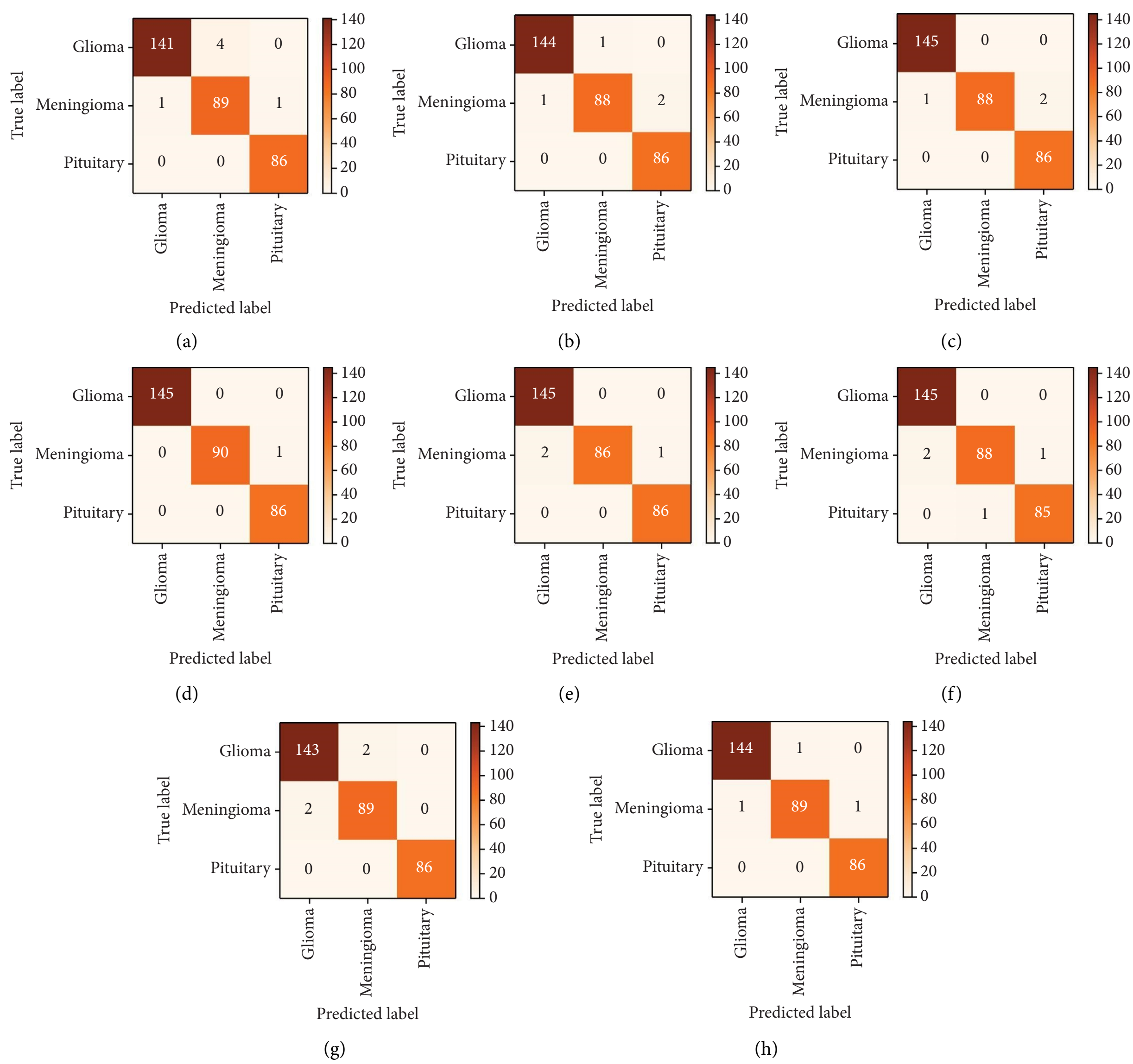


Figure 8: Confusion matrix of EfficientNet models. (a) EfficientNetB0. (b) EfficientNetB1. (c) EfficientNetB2. (d) EfficientNetB3. (e) EfficientNetB4. (f) EfficientNetB5. (g) EfficientNetB6. (h) EfficientNetB7.

In conclusion, based on our analysis, EfficientNetB3 outperforms the other models in terms of accuracy for brain cancer detection. Its superior performance can be attributed to its architectural complexity, capacity for representation, effective generalization, and fine-tuning. This result highlights the importance of selecting an appropriate DL architecture when tackling specific (brain) medical image classification tasks.

4.7. Model Training Process. Our model training process involves several key steps to ensure robust and accurate brain tumor classification:

(1) Data preparation:

  (i) Image preprocessing: We resize all MRI images to 224×224 pixels, reorder the channels to match the RGB format, apply image filtering to enhance clarity, and scale pixel values for numerical stability. Each image is labeled according to the tumor type it represents (meningioma, glioma, or pituitary tumor).

  (ii) Data augmentation: To increase the diversity of the training data and prevent overfitting, we apply various augmentation techniques such as random rotations, flips, zooms, and contrast adjustments.

(2) Transfer learning with EfficientNet:

  (i) We utilize pretrained EfficientNet models (B0 to B7), which are fine-tuned on our specific dataset. This approach leverages the advanced feature extraction capabilities of EfficientNet models, which have been pretrained on a large and diverse image dataset.

(3) Fine-tuning:

(i) The entire EfficientNet model is set to be trainable, allowing us to fine-tune all layers. This enables the model to adapt specifically to the characteristics of our brain tumor MRI images.

(4) Optimization and training:

(i) We use the Adam optimizer with a learning rate of 0.0001 and sparse categorical cross-entropy as the loss function. The model is trained using a batch size of 32, and early stopping is implemented to prevent overfitting.

4.8. Model Selection Criteria. To choose the best model, we evaluate the performance of each EfficientNet variant (B0 to B7) based on several metrics:

(i) Accuracy: The primary metric for assessing the model's ability to correctly classify brain tumor types.

(ii) Loss: Monitoring the loss function during training and validation to ensure that the model is learning effectively.

(iii) Precision, recall, F1-score, sensitivity, specificity, AUC-ROC, confusion matrix, MAE, MSE, and RMSE: These metrics provide additional insights into the model's performance, especially for imbalanced datasets.

4.9. Hyperparameter Analysis. The effects of various hyperparameters on model performance are analyzed as follows:

(1) Learning rate:

(i) We experimented with different learning rates (e.g., 0.001, 0.0001, and 0.00001) to find the optimal rate that ensures stable and efficient learning without causing the model to converge too quickly or get stuck in local minima. We found that the learning rate of 0.0001 provides better performance.

(2) Batch size:

(i) Different batch sizes (e.g., 16, 32, 64) were tested to observe their impact on model convergence and stability. A batch size of 32 was found to provide a good balance between computational efficiency and model performance.

(3) Data augmentation techniques:

(i) The application of data augmentation techniques was carefully tuned. For instance, varying the degree of random rotations, zoom levels, and contrast adjustments helped in understanding their influence on model generalization.

(4) Model depth and width:

(i) We compared the performance of different EfficientNet models (B0 to B7), each with varying depths and widths. This comparison allowed us to identify the model that offered the best trade-off between computational efficiency and classification accuracy.

By systematically analyzing and tuning these hyperparameters, we ensured that the selected EfficientNet model was optimized for our brain tumor classification task. The final chosen model, EfficientNet-B3, demonstrated high accuracy and robustness, making it suitable for practical medical diagnostic applications.

5. Discussion

Table 2 presents a comparison analysis of various DL models for brain tumor classification. In the comparison, eight different research efforts are listed, each associated with a specific model or technique, and all employing the same dataset of T1-weighted CE-MRI images, with a uniform count of 3,064 images across the studies.

Talukder et al. [18] achieved a near-perfect accuracy score of 99.68% using the ResNet50V2 model, which is a variant of the widely recognized ResNet architecture. Notably, the accuracy scores vary among the different models, with Swati et al. [39] reporting an accuracy of 94.82% using a CNN and Huang et al. [30] achieving 95.49% with their CNNBCN model. Ayadi et al. [41] also used a CNN to obtain a 94.74% accuracy rate, while Paul et al. [42] reported a 93% accuracy score with their CNN approach. Belaid et al. [29] utilized the VGG-16 model, a CNN architecture known for its deep structure, and reported a 96.50% accuracy. Sejuti et al. [31] combined CNN with an SVM for classification, achieving a 97.1% accuracy score. Aamir et al. [45] got 98.98% accuracy using adaptive fusion network, Aamir et al. [46] used PLS + clustering and received 98.95%, besides Guan et al. [47] utilized clustering + neural network and got 98.04% in brain tumor classification. Lastly, the comparison includes "Our Proposal," which implies a study or model proposition by the authors of the current document, employing EfficientNetB3 and surpassing others with a 99.69% accuracy score for accurate brain tumor classification. The comparison underscores the ongoing advancements in DL techniques for medical image analysis, particularly in brain tumor classification, highlighting the effectiveness of different architectures and the progress toward highly accurate models. Our proposal leverages the EfficientNetB3 architecture, which is specifically optimized for efficiency, enabling it to achieve a higher accuracy of 99.69% by effectively balancing model complexity and depth. The fine-tuning process is meticulously tailored to the nuances of the T1-weighted CE-MRI brain tumor images, enhancing the model's capability to discern subtle variations within the data. Additionally, our use of advanced data

Table 2: Comparison of brain tumor classification models.

SI. no	Authors	Model/technique	Dataset	No. of images	Accuracy score (%)
1	Talukder et al. [18]	ResNet50V2	T1-weighted CE-MRI	3064	99.68
2	Belaid and Loudini [29]	VGG-16	T1-weighted CE-MRI	3064	96.50
3	Huang et al. [30]	CNNBCN	T1-weighted CE-MRI	3064	95.49
4	Sejuti and Islam [31]	CNN-SVM	T1-weighted CE-MRI	3064	97.10
5	Swati et al. [39]	CNN	T1-weighted CE-MRI	3064	94.82
6	Ayadi et al. [41]	CNN	T1-weighted CE-MRI	3064	94.74
7	Paul et al. [42]	CNN	T1-weighted CE-MRI	3064	93.00
8	Aaamir et al. [45]	Adaptive fusion network	T1-weighted CE-MRI	3064	98.98
9	Aaamir et al. [46]	PLS + clustering	T1-weighted CE-MRI	3064	98.95
10	Guan et al. [47]	Clustering + neural network	T1-weighted CE-MRI	3064	98.04
11	Qureshi et al. [51]	UL-DLA + GLCM	T1-weighted CE-MRI	3064	99.23
12	Our proposal	EfficientNetB3	T1-weighted CE-MRI	3064	99.69

augmentation techniques and regularization strategies effectively combats overfitting, further boosting the model's performance on the dataset compared to other approaches.

The distinguishing qualities of our research were compared to those of the previous studies in the field of deep learning for brain tumor classification. Our study stands out in several aspects:

(1) Innovative architecture selection: Unlike many previous studies that utilize well-established deep learning architectures such as ResNet, CNN, or VGG, we adopt the EfficientNet architecture. This choice is motivated by EfficientNet's superior performance and efficiency, which is achieved through a unique compound scaling method, making it particularly well suited for medical image analysis tasks like brain tumor classification.

(2) Tailored transfer learning and fine-tuning: We employ a meticulous approach to transfer learning, reconfiguring the EfficientNet architecture and fine-tuning it specifically for brain tumor classification. This involves adding and modifying layers to adapt the model to the intricacies of brain MRI data, optimizing its performance for this particular task.

(3) Comprehensive preprocessing and augmentation: Our research incorporates extensive preprocessing steps, including intensity normalization and contrast enhancement, which are crucial for improving the quality and consistency of the input data. Additionally, we implement sophisticated image augmentation techniques to mitigate overfitting and enhance the model's generalization ability, a facet not extensively explored in some prior studies.

(4) Immediate image standardization: A unique aspect of our methodology is the immediate image standardization step in the augmentation process. This ensures uniformity in the input data, reducing variance and enhancing the model's ability to learn effectively from augmented images.

(5) Thorough evaluation and comparison: We conduct a comprehensive evaluation of our proposed models using various performance metrics, including accuracy, precision, recall, F1-score, sensitivity, specificity, AUC-ROC, confusion matrix, MAE, MSE, and RMSE. This allows for a rigorous comparison with previous studies, demonstrating the effectiveness and superiority of our approach.

In summary, our research distinguishes itself through the strategic selection of architecture, meticulous transfer learning and fine-tuning, comprehensive preprocessing and augmentation, immediate image standardization, and thorough evaluation metrics. These unique aspects collectively contribute to the superior performance and reliability of our methodology compared to previous studies in the field. We believe that these contributions significantly advance the state-of-the-art in deep learning-based brain tumor classification and hold promise for improving clinical diagnostics and patient outcomes.

5.1. Differentiation between EfficientNet Models. Each member of the EfficientNet family, from B0 to B7, represents a progression in terms of depth, width, and overall capacity. While the core architecture remains consistent across all variants, each model incorporates specific enhancements or modifications to address the unique challenges of image classification tasks, including brain tumor classification.

EfficientNetB3, in particular, stands out due to several key adjustments tailored to optimize its performance for brain tumor classification:

(i) Increased depth and width: EfficientNetB3 features a deeper and wider architecture compared to its predecessors (B0, B1, B2). This increase in depth and width allows B3 to capture more intricate patterns and features in brain MRI images, enhancing its ability to distinguish between different tumor types with greater accuracy.

(ii) Enhanced representation capacity: With its larger capacity for representing image features, EfficientNetB3 can effectively differentiate between benign and malignant tumors. This enhanced representation capacity is crucial for accurately classifying brain tumors, which often exhibit subtle variations in appearance and texture.

(iii) Fine-tuning and adaptation: EfficientNetB3 benefits from fine-tuning and adaptation processes specifically tailored to the nuances of brain tumor classification. Fine-tuning allows the model to specialize in this task by adjusting its parameters to better fit the characteristics of brain MRI images, leading to improved accuracy and performance.

(iv) Effective generalization: EfficientNetB3's architecture demonstrates effective generalization to the complexities of brain tumor classification, striking a balance between complexity and generalization. This optimal balance enables B3 to perform well on a wide range of brain MRI datasets, making it a versatile and reliable choice for this critical medical image analysis task.

5.2. Insight into EfficientNetB3's Superior Performance. The superior performance of EfficientNetB3 for brain tumor classification can be attributed to its unique adjustments, including increased depth and width, enhanced representation capacity, fine-tuning and adaptation processes, and effective generalization. These modifications enable B3 to capture subtle variations in brain MRI images, accurately distinguishing between different tumor types and contributing to its exceptional accuracy and reliability.

In conclusion, EfficientNetB3's specific enhancements and modifications make it ideally suited for brain tumor classification, leading to its superior performance compared to other EfficientNet variants. By leveraging these unique adjustments, we can develop more robust and accurate brain tumor classification systems, ultimately benefiting patient care and medical research.

5.3. Computational Efficiency and Processing Times. Our model, based on EfficientNet architecture, is designed to strike a balance between computational efficiency and accuracy. The EfficientNet models are known for their compact yet powerful design, making them suitable for resource-constrained environments such as healthcare facilities.

(i) Processing times: The processing times for image analysis vary depending on factors such as the size of the input images, the complexity of the model architecture, and the computational resources available. However, our experiments have shown that the EfficientNet models can perform image classification tasks with relatively low inference times, typically ranging from milliseconds to a few seconds per image on modern hardware.

(ii) Hardware requirements: Deploying our model in real-world diagnostic workflows requires hardware capable of running deep learning algorithms efficiently. While the exact hardware requirements may vary depending on the specific model variant and the size of the dataset, a modern GPU (Graphics Processing Unit) or a dedicated AI accelerator can significantly accelerate the inference process.

5.4. Deployment Considerations

(i) Integration with existing infrastructure: Our model can be seamlessly integrated into existing healthcare IT systems, allowing for streamlined diagnostic workflows. Integration with Picture Archiving and Communication Systems (PACS) and Electronic Health Records (EHR) platforms ensures interoperability and data consistency.

(ii) Scalability: The scalability of our model allows it to handle large volumes of imaging data efficiently, making it suitable for deployment in high-throughput clinical environments.

(iii) Continued optimization: We are committed to ongoing optimization efforts to further improve the model's computational efficiency and reduce processing times. This includes exploring techniques such as model pruning, quantization, and hardware acceleration to optimize resource utilization and enhance real-time performance.

In conclusion, our model prioritizes computational efficiency to enable timely and accurate diagnoses in clinical settings. By detailing processing times, hardware requirements, and deployment considerations, we aim to facilitate the seamless integration of our model into real-world diagnostic workflows, ultimately benefiting healthcare professionals and patients alike.

5.5. Integrating the Proposed Model into Healthcare IT Systems

(1) Data privacy: To ensure data privacy, our model integration will comply with regulations such as HIPAA and GDPR. We will anonymize MRI images, use secure encryption for data transfer and storage, implement access control mechanisms, and conduct regular audits to maintain privacy standards.

(2) Model interpretability: We will enhance model interpretability using tools like Grad-CAM and LIME, which provide visual and textual explanations of the model's decisions. This transparency will help clinicians understand the model's rationale, fostering trust and aiding in clinical decision-making.

(3) User interface design: Our user-friendly interface will feature a dashboard displaying MRI images, model predictions, and explanations. It will support customizable report generation and integration with electronic health record (EHR) systems. Designed with clinician input, the interface will ensure intuitive navigation and efficient use in clinical settings, supported by user training and materials.

The feasibility of integrating our brain tumor classification model into existing healthcare IT systems, emphasizing data privacy, model interpretability, and user interface design to ensure seamless and effective clinical adoption.

5.6. Potential Challenges in Clinical Implementation

(1) Ongoing model training: One potential challenge in clinical implementation is the need for ongoing model training with new data. As new MRI data and updated medical knowledge become available, our model must be periodically retrained to maintain its accuracy and relevance. This process involves

(i) Continuous data collection: Establishing protocols for the regular collection of new MRI scans and diagnostic data.
(ii) Incremental learning: Implementing incremental learning techniques to update the model without requiring complete retraining from scratch, thereby reducing computational costs and time.
(iii) Validation and monitoring: Regularly validating the updated model against new and diverse datasets to ensure its robustness and generalizability. Continuous performance monitoring will help identify any degradation in model accuracy over time.

(2) Integration with Electronic Health Records (EHR) Systems: Integrating the model with existing EHR systems presents another challenge due to the complexity and diversity of these systems. Key strategies to address this include

(i) Standardized data formats: Using standardized data formats (e.g., HL7 and FHIR) to facilitate seamless data exchange between the model and EHR systems.
(ii) API development: Developing robust APIs that enable smooth communication between the model and various EHR platforms, ensuring that patient data can be efficiently retrieved and updated.
(iii) Interoperability testing: Conducting extensive interoperability testing with different EHR systems to identify and resolve compatibility issues, ensuring the model's integration is smooth and effective across various clinical environments.

By addressing these challenges, we aim to ensure that our brain tumor classification model can be effectively and sustainably implemented in clinical settings, enhancing diagnostic capabilities and improving patient outcomes.

5.7. Roadmap for Extending Model Applications

5.7.1. Expansion to Other Tumor Types

(i) Objective: Broaden the scope of the model to classify other tumor types, such as lung, breast, and liver tumors.
(ii) Approach:

(i) Data collection: Gather diverse datasets for different tumor types to ensure the model is trained on a wide range of images.
(ii) Model adaptation: Fine-tune the EfficientNet architecture to accommodate the specific features and characteristics of different tumors.
(iii) Validation: Conduct rigorous testing and validation to ensure the model's accuracy and robustness across various tumor types.

5.7.2. Integration of Multiple Imaging Modalities

(i) Objective: Enhance the model's diagnostic capabilities by incorporating different imaging techniques such as CT scans, PET scans, and ultrasound images.
(ii) Approach:

(i) Multimodal data fusion: Develop algorithms to integrate data from multiple imaging modalities, leveraging their complementary strengths.
(ii) Hybrid models: Explore hybrid approaches that combine information from different imaging sources to improve diagnostic accuracy.
(iii) Cross-training: Train the model on multimodal datasets to ensure it can effectively handle and interpret data from various imaging techniques.

By extending our model to other tumor types and imaging modalities, we aim to create a versatile and comprehensive diagnostic tool that can significantly improve early detection and treatment across various medical conditions. This roadmap underscores our commitment to advancing medical AI and providing robust, accurate, and accessible diagnostic solutions.

5.8. Limitations Related to Dataset Diversity. Addressing the limitations related to dataset diversity is crucial for the robustness and generalizability of deep learning models, particularly in medical imaging, where variations in age, tumor stages, and imaging techniques can significantly impact performance. If the dataset predominantly comprises images from a specific age group, such as adults, it may not adequately capture the nuances and variations present in pediatric or geriatric populations, affecting the model's ability to accurately detect and classify tumors across different age groups. Moreover, an imbalanced representation of tumor stages, with a majority of images from early stages and fewer images from advanced stages, could skew the model's performance towards the more represented classes, compromising its ability to generalize to cases with less representation, such as advanced tumors. Additionally, the diversity of imaging techniques used to capture tumor images can impact the model's generalizability; for instance, if the dataset predominantly consists of images obtained from a single imaging modality, such as MRI, while other modalities like CT scans or ultrasound are underrepresented, the model may struggle to generalize its findings across different imaging modalities. Future efforts should focus on enhancing dataset diversity by including a broader range of age groups, tumor stages, and imaging modalities to develop more robust and widely applicable models [52–56].

6. Conclusion

This study presents a novel approach to brain tumor classification, leveraging the advanced capabilities of the EfficientNet family of deep learning models. Accurate and early detection of brain tumors is crucial for improving patient outcomes, and our work aims to enhance this diagnostic process through sophisticated deep learning techniques. We conducted a comprehensive analysis using MRI data from the publicly accessible Figshare dataset, employing various iterations of the EfficientNet architecture, from EfficientNetB0 to EfficientNetB7. Each model's performance was rigorously evaluated, with a primary focus on accuracy. The results demonstrate that the EfficientNet models achieve high accuracy rates, with EfficientNetB0 at 98.14%, EfficientNetB1 at 98.76%, EfficientNetB2 at 99.07%, EfficientNetB3 at 99.69%, EfficientNetB4 at 99.07%, EfficientNetB5 at 98.76%, EfficientNetB6 at 98.76%, and EfficientNetB7 at 99.07%. Notably, the EfficientNetB3 model achieved the highest accuracy at 99.69%, indicating its superior performance in brain tumor classification tasks. The exceptional accuracy of the EfficientNetB3 model represents a significant advancement in medical diagnostics, particularly for the early detection and classification of brain tumors. Implementing such a model in clinical settings can greatly enhance diagnostic reliability, facilitate timely and effective interventions, and ultimately improve patient care.

While our study demonstrated the efficacy of EfficientNet models for brain tumor classification, it is not without limitations. The reliance on publicly available datasets, such as the Figshare dataset, may not fully capture the diversity of real-world clinical scenarios. Moreover, our focus on EfficientNet models alone, without exploring alternative approaches like mixture models or feature fusion techniques, represents another constraint that may also limit their practical application in resource-constrained clinical environments.

To address these limitations, future research will focus on several initiatives. We plan to collaborate with healthcare institutions to obtain more diverse and representative MRI datasets. Exploring hybrid approaches that integrate multiple data types, such as combining MRI images with clinical data, could enhance model robustness and accuracy. We will also investigate feature fusion techniques to leverage complementary information from various sources to make advanced models in clinical practice.

Data Availability

The selected datasets are sourced from free and open-access sources such as The Figshare Brain Tumor dataset https://figshare.com/articles/dataset/brain_tumor_dataset/1512427.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Authors' Contributions

Md. Manowarul Islam conducted conceptualization, methodology, visualization, investigation, supervision, and writing-original draft preparation; Md. Alamin Talukder conducted conceptualization, data curation, methodology, resource, software, visualization, formal analysis, writing, reviewing, and editing; Md Ashraf Uddin and Arnisha Akhter conducted formal analysis, investigation, validation, writing, reviewing, and editing; Majdi Khalid conducted investigation, validation, writing, reviewing, and editing;

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